Effects of Conditional Cash Transfers on School Dropout Rates: A Bayesian Approach to Modelling Linear Regression

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Social program on education: Prospera

- 1988, implementation of social program, Prospera.
- Purpose: improve education outcomes in the country.
- "Solidarity, Progress, Well-being..."
  - **Prospera**
- Works through Conditional Cash Transfers provided to mothers.
  - Conditional on school attendance
- Goal of this analysis: Bayesian approach to the linear regression by computing posterior distributions of θ.
  - Interested in treatment effect of receiving cash transfers
- **Data:** 79,984 units at school level, complete population of the most marginalized schools, according to the Mexican Population Council.

Model specification

- The linear model considered is:

$$\begin{aligned} Y_i = \beta_0 &+ \beta_1 \,\mathsf{treatment}_i + \beta_2 \,\mathsf{nteacher}_i \\ &+ \beta_3 \,\mathsf{nteacher}_i^2 + \beta_4 \,\mathsf{teacherbcs}_i \\ &+ \beta_5 \,\mathsf{csize}_i + \epsilon_i \end{aligned} \tag{1}$$

- Parameters $\theta = (\boldsymbol{\beta}, \sigma^2)$

Priors

- Priors implemented are:

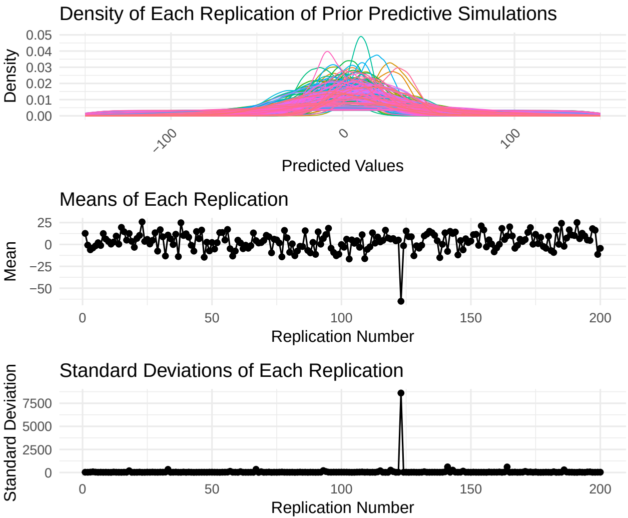
$$\beta \sim N_p(\beta_0, \Sigma_0) \tag{2}$$

and

$$\sigma^2 \sim IG\left(\frac{\nu_0}{2}, \frac{\nu_0 \sigma_0^2}{2}\right). \tag{3}$$

- $\boldsymbol{\beta_0} = \hat{\boldsymbol{\beta}}_{\boldsymbol{ols}}$ and $\Sigma_0 = \frac{\boldsymbol{X'X}}{n\hat{\sigma_{ols}}^2}$
- Uninformativeness due to:
  - Objectiveness due to program evaluation
  - Studies vary a lot
  - Complexity of model specification

Prior predictive check



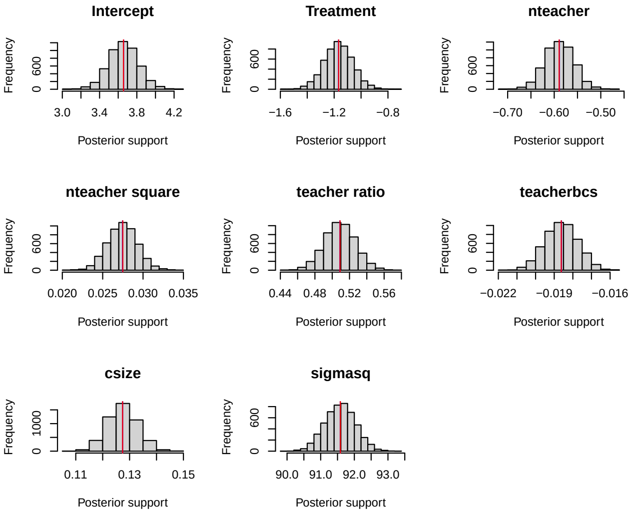
Computing posteriors

- Y_i is modeled by:

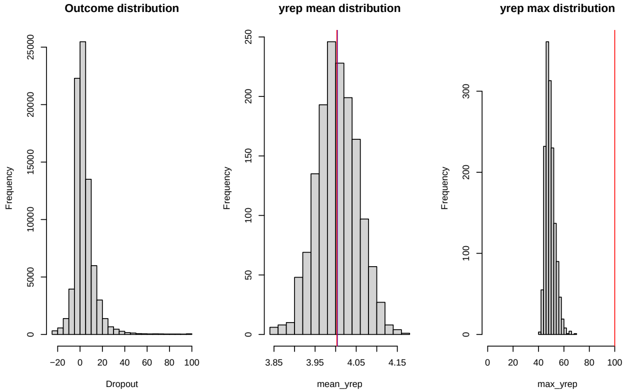
$$\{Y \mid \mathbf{X}, \boldsymbol{\beta}, \sigma^2\} \sim \text{multivariate normal}(\mathbf{X}\boldsymbol{\beta}, \sigma^2\mathbf{I})$$

- Rely on Gibbs sampling MCMC method to compute posterior distributions for θ.

Posteriors


Intercept
Frequency
0
600
3.0
3.4
3.8
4.2
Posterior support
Treatment
Frequency
0
600
−1.6
−1.2
−0.8
Posterior support
nteacher
Frequency
0
600
−0.70
−0.60
−0.50
Posterior support
nteacher square
Frequency
0
600
0.020
0.025
0.030
0.035
Posterior support
teacher ratio
Frequency
0
600
0.44
0.48
0.52
0.56
Posterior support
teacherbcs
Frequency
0
600
−0.022
−0.019
−0.016
Posterior support
csize
Frequency
0
1000
0.11
0.13
0.15
Posterior support
sigmasq
Frequency
0
600
90.0
91.0
92.0
93.0
Posterior support

Posterior predictive check



- $T(y)_{obs} = \sum_{i=1}^{n} y_i / n$ where $p_B = 0.5006667$
- $T(y) = max|y_i|$ where $p_B = 0$

Results

Table: Posterior Means and 95% Credible Intervals for Betas and σ^2

	Posterior Mean	2.5%	97.5%
(Intercept)	3.6576	3.3366	3.9713
Cash transfer	-1.1674	-1.3723	-0.9507
Num. teachers	-0.5890	-0.6498	-0.5270
Num. teachers^2	0.0275	0.0238	0.0311
Teacher ratio	0.5089	0.4740	0.5439
Teacher with bc's	-0.0186	-0.0203	-0.0170
Class size	0.1274	0.1163	0.1384
σ^2	91.59279	90.70267	92.46518